



# ROTARY DOUBLE INVERTED PENDULUM MODELING & CONTROL

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## Undergraduate Research

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### Abstract

This report presented the mathematical modeling and control design of a rotary double inverted pendulum. A nonlinear model was derived using Lagrangian mechanics and linearized about the upright equilibrium. State-space methods were then used to design stabilizing and tracking controllers, whose performance was evaluated through simulation and experimental testing.

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## 1 Nomenclature

$\theta$  Rotary arm angular position [rad]

$\alpha$  First pendulum angular position [rad]

$\phi$  Second pendulum angular position [rad]

$\dot{\theta}$  Rotary arm angular velocity [rad/s]

$\dot{\alpha}$  First pendulum angular velocity [rad/s]

$\dot{\phi}$  Second pendulum angular velocity [rad/s]

$\ddot{\theta}$  Rotary arm angular acceleration [rad/s<sup>2</sup>]

$\ddot{\alpha}$  First pendulum angular acceleration [rad/s<sup>2</sup>]

$\ddot{\phi}$  Second pendulum angular acceleration [rad/s<sup>2</sup>]

$V_m$  Input motor voltage [V]

$r$  Rotary arm length [m]

$l_1$  Distance from the first pendulum joint to its centre of mass [m]

$l_2$  Distance from the second pendulum joint to its centre of mass [m]

$L_r$  Total length of the rotary arm [m]

$L_1$  Total length of the first pendulum [m]

$L_2$  Total length of the second pendulum [m]

$m_r$  Mass of the rotary arm [kg]

$m_1$  Mass of the first pendulum [kg]

$m_h$  Mass of the hinge [kg]

$m_2$  Mass of the second pendulum [kg]

$J_r$  Mass moment of inertia of the rotary arm about the fixed frame [kg·m<sup>2</sup>]

$J_1$  Mass moment of inertia of the first pendulum about its centre of mass [kg·m<sup>2</sup>]

$J_2$  Mass moment of inertia of the second pendulum about its centre of mass [kg·m<sup>2</sup>]

$g$  Gravitational acceleration [m/s<sup>2</sup>]

$\tau$  Applied torque at the rotary arm [N·m]

## 2 DBIP Diagram - Quanser

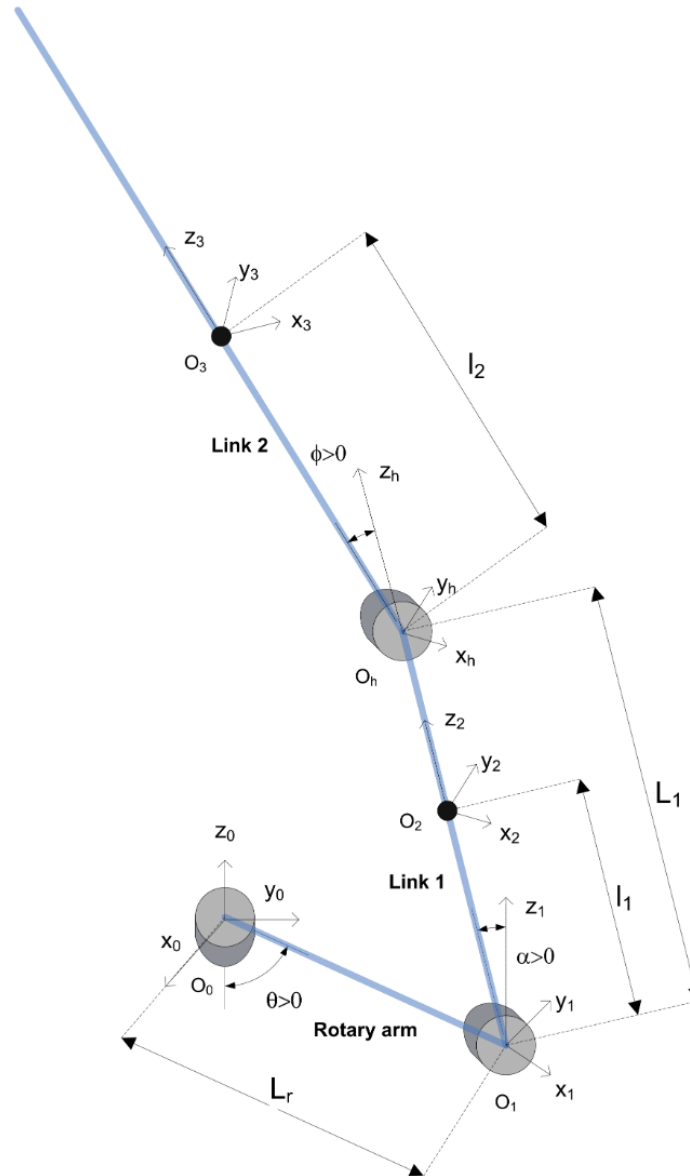


Figure 1: Double Inverted Pendulum

### 3 Problem Statement & Definitions

In the following modeling of the system, it's assumed the masses are concentrated about the center of mass of the respective geometries and that they are rigid bodies. The frames of the double inverted pendulum (DBIP) are defined as follows:

- Frame 0 : Inertial Frame
- Frame 1 : Tip of the Arm
- Frame 2: Tip of the Bottom Link (Hinge)
- Frame 3: Tip of the Top Link

#### 3.1 Rotation Matrices

The following section presents rotation matrices about the x and z axis that are used to relate the different frames of reference.

**Frame 1 relative to Frame 0:**

$$\mathbf{R}_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

**Frame 2 relative to Frame 1:**

$$\mathbf{R}_2^1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

**Frame 3 relative to Frame 2:**

$$\mathbf{R}_3^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix}$$

#### 3.2 Position Vectors

The following relates the resultant position vectors of the tip of the arm, center of mass of link 1, the hinge, the center of mass of link 2 relative to the inertial frame.

$$\mathbf{r}_{arm}^0 = \begin{bmatrix} L_r \cos(\theta) \\ L_r \sin(\theta) \\ 0 \end{bmatrix} \quad (1)$$

$$\mathbf{r}_1^0 = \begin{bmatrix} L_r \cos(\theta) + l_1 \sin(\alpha) \sin(\theta) \\ L_r \sin(\theta) - l_1 \sin(\alpha) \cos(\theta) \\ l_1 \cos(\alpha) \end{bmatrix} \quad (2)$$

$$\mathbf{r}_{hinge}^0 = \begin{bmatrix} L_r \cos(\theta) + L_1 \sin(\alpha) \sin(\theta) \\ L_r \sin(\theta) - L_1 \sin(\alpha) \cos(\theta) \\ L_1 \cos(\alpha) \end{bmatrix} \quad (3)$$

$$\mathbf{r}_2^0 = \begin{bmatrix} \sin(\theta) (L_1 \sin(\alpha) + l_2 \cos(\alpha) \sin(\phi) + l_2 \cos(\phi) \sin(\alpha)) + L_r \cos(\theta) \\ L_r \sin(\theta) - \cos(\theta) (L_1 \sin(\alpha) + l_2 \cos(\alpha) \sin(\phi) + l_2 \cos(\phi) \sin(\alpha)) \\ L_1 \cos(\alpha) + l_2 \cos(\alpha) \cos(\phi) - l_2 \sin(\alpha) \sin(\phi) \end{bmatrix} \quad (4)$$

### 3.3 Velocity Vectors

The velocity vectors are obtained by applying the Jacobian (first-order Taylor expansion of a multivariable function )

$$\dot{\mathbf{r}} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \dot{\mathbf{q}} \quad (5)$$

where

$$\mathbf{r} = \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} \theta \\ \alpha \\ \phi \end{bmatrix} \quad (6)$$

$$\dot{\mathbf{r}} = \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \\ \dot{r}_z \end{bmatrix} = \begin{bmatrix} \frac{\partial r_x}{\partial \theta} & \frac{\partial r_x}{\partial \alpha} & \frac{\partial r_x}{\partial \phi} \\ \frac{\partial r_y}{\partial \theta} & \frac{\partial r_y}{\partial \alpha} & \frac{\partial r_y}{\partial \phi} \\ \frac{\partial r_z}{\partial \theta} & \frac{\partial r_z}{\partial \alpha} & \frac{\partial r_z}{\partial \phi} \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix}$$

The derived velocity vectors are:

$$\dot{\mathbf{r}}_{arm}^0 = \begin{bmatrix} -L_r \sin(\theta) & 0 & 0 \\ L_r \cos(\theta) & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} -L_r \dot{\theta} \sin(\theta) \\ L_r \dot{\theta} \cos(\theta) \\ 0 \end{bmatrix} \quad (7)$$

$$\dot{\mathbf{r}}_1^0 = \begin{bmatrix} l_1 \sin(\alpha) \cos(\theta) - L_r \sin(\theta) & l_1 \cos(\alpha) \sin(\theta) & 0 \\ L_r \cos(\theta) + l_1 \sin(\alpha) \sin(\theta) & -l_1 \cos(\alpha) \cos(\theta) & 0 \\ 0 & -l_1 \sin(\alpha) & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} \quad (8)$$

$$\begin{aligned}
&= \begin{bmatrix} \dot{\alpha} l_1 \cos(\alpha) \sin(\theta) - \dot{\theta} (L_r \sin(\theta) - l_1 \sin(\alpha) \cos(\theta)) \\ \dot{\theta} (L_r \cos(\theta) + l_1 \sin(\alpha) \sin(\theta)) - \dot{\alpha} l_1 \cos(\alpha) \cos(\theta) \\ -\dot{\alpha} l_1 \sin(\alpha) \end{bmatrix} \\
\dot{\mathbf{r}}_h^0 &= \begin{bmatrix} L_1 \sin(\alpha) \cos(\theta) - L_r \sin(\theta) & L_1 \cos(\alpha) \sin(\theta) & 0 \\ L_r \cos(\theta) + L_1 \sin(\alpha) \sin(\theta) & -L_1 \cos(\alpha) \cos(\theta) & 0 \\ 0 & -L_1 \sin(\alpha) & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} \quad (9)
\end{aligned}$$

$$= \begin{bmatrix} L_1 \dot{\alpha} \cos(\alpha) \sin(\theta) - \dot{\theta} (L_r \sin(\theta) - L_1 \sin(\alpha) \cos(\theta)) \\ \dot{\theta} (L_r \cos(\theta) + L_1 \sin(\alpha) \sin(\theta)) - L_1 \dot{\alpha} \cos(\alpha) \cos(\theta) \\ -L_1 \dot{\alpha} \sin(\alpha) \end{bmatrix}$$

$$\dot{\mathbf{r}}_2^0 = \mathbf{J}_2(\mathbf{q})\dot{\mathbf{q}} = \begin{bmatrix} \dot{r}_x \\ \dot{r}_y \\ \dot{r}_z \end{bmatrix}, \quad (10)$$

## 4 Potential Energy of the System

The potential energies of the bottom link, hinge and the top link are respectively defined as:

$$V_1 = m_1 g r_{1z}^0 \quad (11)$$

$$V_h = m_h g r_{hz}^0 \quad (12)$$

$$V_2 = m_2 g r_{2z}^0 \quad (13)$$

$$V_t = V_1 + V_h + V_2 \quad (14)$$

### 4.1 Total Potential Energy

The resulting total potential energy is given symbolically by:

$$V_t = g m_2 (l_2 \cos(\alpha + \phi) + L_1 \cos(\alpha)) + L_1 g m_h \cos(\alpha) + g l_1 m_1 \cos(\alpha) \quad (15)$$

## 5 Kinetic Energy of the System

This section defines the kinetic energy of the rotary arm, bottom link, hinge and top link. The kinetic energy of a rigid object is the sum of two terms, the translational kinetic energy obtained by concentrating the entire mass of the object at the center of mass, and the rotational kinetic energy of the body about the center of mass

$$KE_{\text{total}} = KE_{\text{trans}} + KE_{\text{rot}} = \frac{1}{2}m\mathbf{v}^T\mathbf{v} + \frac{1}{2}\boldsymbol{\omega}^T\mathcal{J}\boldsymbol{\omega} \quad (16)$$

where  $m$  is the total mass of the rigid body,  $v$  and  $\omega$  are the linear and angular velocity vectors relative to the inertial frame.  $\mathcal{J}$  is the inertia tensor defined as

$$\mathcal{J} = RIR^T \quad (17)$$

where  $I$  is the inertia tensor expressed in the body-attached frame and  $R$  is the orientation transformation from the body-attached frame and the inertial frame.

### 5.1 Addition of Angular Velocities

The addition of angular velocities are important in computing the rotational kinetic energy of the bottom and top link where the resultant angular velocity is due to the relative rotation of several coordinate frames[2]. In the general case, a time-varying rotation matrix  $R$  is defined as

$$\dot{R} = S(\omega(t))R(t) \quad (18)$$

Where  $S(\omega(t))$  is skew symmetric and  $R(t)$  is the angular velocity of the rotating frame relative to the fixed frame at time  $t$ . In the following, the notation  $\omega_{i,j}$  denotes the angular velocity vector that corresponds to the time derivative of the rotation matrix  $R_j^i$ . The derivations [2] leads to the formula:

$$\omega_{0,n}^0 = \omega_{0,1}^0 + R_1^0\omega_{1,2}^1 + R_2^0\omega_{2,3}^2 + R_3^0\omega_{3,4}^3 + \cdots + R_{n-1}^0\omega_{n-1,n}^{n-1} \quad (19)$$

The focus is in  $\omega_{0,2}^0$  and  $\omega_{0,3}^0$ , the angular velocities for the frames aligned with the bottom and top link relative to the inertial frame.

### 5.2 Kinetic Energy of the Arm

$$\begin{aligned} KE_{\text{total}} &= \frac{1}{2}m(\omega \cdot d)^2 + \frac{1}{2}I \cdot \omega^2 \\ &= \frac{1}{2}\omega^2(I + md^2) \end{aligned} \quad (20)$$

Where the right term is the parallel axis theorem for the inertia<sup>1</sup> about the axis of rotation relative to the inertial frame.  $q_1$  refers to  $\omega = \dot{\theta}$ .

<sup>1</sup>The inertia value given by Quanser isn't correct. Check the next sections for the correct value



### 5.3 Translational Kinetic Energy of Link 1

The translational kinetic energy of link 1 is computed as

$$T_{t1} = \frac{1}{2} m_1 \dot{r}_{01}^2 \quad (21)$$

### 5.4 Rotational Kinetic Energy of Link 1

The inertia tensor about the body attached frame is shown as follows.<sup>2</sup>

$$I_{link1} = \begin{bmatrix} \frac{1}{12} m_1 L_1^2 & 0 & 0 \\ 0 & \frac{1}{12} m_1 L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (22)$$

The inertia tensor  $\mathcal{J}$  obtained through the similarity transformation as seen in (17):

$$\mathcal{J}_{link1} = R_2^0 \cdot I_{link1} \cdot R_2^{0T} \quad (23)$$

$$\begin{bmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} \frac{1}{12} m_1 L_1^2 & 0 & 0 \\ 0 & \frac{1}{12} m_1 L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix}^T$$

The angular velocity of the rotating frame 2 relative to the inertial frame 0 using the formula from (19):

$$\omega_{0,2}^0 = \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 \quad (24)$$

$$= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{\alpha} \cos(\theta) \\ \dot{\alpha} \sin(\theta) \\ \dot{\theta} \end{bmatrix} \quad (25)$$

The resulting rotational kinetic energy of the bottom link is given as:

$$\begin{aligned} T_{r1} &= \frac{1}{2} \omega_{0,2}^{0T} \cdot \mathcal{J}_{link1} \cdot \omega_{0,2}^0 \\ &= \frac{L_1^2 m_1}{48} (2\dot{\alpha}^2 - \dot{\theta}^2 \cos(2\alpha) + \dot{\theta}^2) \end{aligned} \quad (26)$$

### 5.5 Kinetic Energy of the Hinge

The translational kinetic energy of the hinge is described as:

$$T_{htrans} = \frac{1}{2} m_h \dot{r}_{0h}^2 \quad (27)$$

The latter doesn't have rotational kinetic energy because it's treated as a point mass.

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<sup>2</sup>Note that there's no  $I_{zz}$  entry because the link is assumed to be a thin rod.

## 5.6 Translational Kinetic Energy of Link 2

The translational kinetic energy of link 2 is computed as

$$T_{t2} = \frac{1}{2} m_2 \dot{r}_2^2 \quad (28)$$

## 5.7 Rotational Kinetic Energy of Link 2

The inertia tensor about the body attached frame is shown as follows.<sup>3</sup> The inertia tensor is:

$$I_{link2} = \begin{bmatrix} \frac{1}{12} m_2 L_1^2 & 0 & 0 \\ 0 & \frac{1}{12} m_2 L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (29)$$

The inertia tensor  $\mathcal{J}$  obtained through the similarity transformation as seen in (17):

$$\mathcal{J}_{link2} = R_3^0 \cdot I_{link2} \cdot R_3^{0T} \quad (30)$$

$$= \begin{bmatrix} \cos(\theta) & -\cos(\alpha + \phi) \sin(\theta) & \sin(\alpha + \phi) \sin(\theta) \\ \sin(\theta) & \cos(\alpha + \phi) \cos(\theta) & -\sin(\alpha + \phi) \cos(\theta) \\ 0 & \sin(\alpha + \phi) & \cos(\alpha + \phi) \end{bmatrix} \begin{bmatrix} \frac{1}{12} m_2 L_1^2 & 0 & 0 \\ 0 & \frac{1}{12} m_2 L_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\cos(\alpha + \phi) \sin(\theta) & \sin(\alpha + \phi) \sin(\theta) \\ \sin(\theta) & \cos(\alpha + \phi) \cos(\theta) & -\sin(\alpha + \phi) \cos(\theta) \\ 0 & \sin(\alpha + \phi) & \cos(\alpha + \phi) \end{bmatrix}^T$$

The angular velocity of the rotating frame 3 relative to the inertial frame 0 using the formula from (19):

$$\omega_{0,3}^0 = \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 + R_2^0 \omega_{2,3}^2 \quad (31)$$

$$= \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \cos(\theta) & -\cos(\alpha) \sin(\theta) & \sin(\alpha) \sin(\theta) \\ \sin(\theta) & \cos(\alpha) \cos(\theta) & -\sin(\alpha) \cos(\theta) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos(\theta)(\dot{\alpha} + \dot{\phi}) \\ \sin(\theta)(\dot{\alpha} + \dot{\phi}) \\ \dot{\theta} \end{bmatrix}$$

The resulting rotational kinetic energy of the top link is given as:

$$T_{r2} = \frac{1}{2} \cdot \omega_{0,3}^{0T} \cdot \mathcal{J}_{link2} \cdot \omega_{0,3}^0 \quad (32)$$

$$= \frac{L_1^2 m_2}{48} (4\dot{\alpha}\dot{\phi} - \dot{\theta}^2 \cos(2\alpha + 2\phi) + 2\dot{\alpha}^2 + 2\dot{\phi}^2 + \dot{\theta}^2)$$

## 5.8 Sum of Kinetic Energy

The resulting total kinetic energy is given symbolically by:

$$T_t = T_{arm} + T_{1trans} + T_{r1} + T_{htrans} + T_{2trans} + T_{r2} \quad (33)$$

---

<sup>3</sup>Note that there's no  $I_{zz}$  entry because it's assumed to be a thin rod.

## 6 Lagrangian Formulation

Using MATLAB symbolic toolkit, the following Lagrange equation is given as:

$$L(q, \dot{q}) = T_t - V_t \quad (34)$$

### 6.1 Generalized Forces

The generalized forces include the torque and the damping coefficients:

$$\tau = \begin{bmatrix} \tau_1 - D_r \dot{\theta} \\ -D_1 \dot{\alpha} \\ -D_2 \dot{\phi} \end{bmatrix} \quad (35)$$

where

$$\tau = \frac{\eta_g K_g \eta_m k_t (V_m - K_g k_m \dot{\theta})}{R_m} \quad (36)$$

### 6.2 Euler Lagrange Equation

The Euler Lagrange Equation is of the form:

$$\frac{\partial^2 L}{\partial t \partial \dot{q}} - \frac{\partial L}{\partial q} = Q \quad (37)$$

MATLAB computes the following symbolically:

$$\frac{\partial^2 L}{\partial t \partial \dot{q}} - \frac{\partial L}{\partial q} - Q_i = 0$$

and solve for  $\ddot{\theta}, \ddot{\alpha}, \ddot{\phi}$ . The final nonlinear model has the form:

$$\dot{x} = f(x, u) = [\dot{\theta}, \dot{\alpha}, \dot{\phi}, \ddot{\theta}, \ddot{\alpha}, \ddot{\phi}]^T \quad (38)$$

with the state vector:

$$x = [\theta, \alpha, \phi, \dot{\theta}, \dot{\alpha}, \dot{\phi}]^T \quad (39)$$

### 6.3 Linearization - ABCD Realization

The controlled equilibrium point is the zero  $x$  vector with no voltage input. By computing the Jacobians:

$$x_{eq} = [\theta \ 0 \ 0 \ 0 \ 0 \ 0]^T \quad \text{and} \quad v_{eq} = 0 \quad (40)$$

It should be noted that several inconsistencies were identified in the Quanser documentation, including a discrepancy in the reported inertia value of the rotary arm and inconsistencies in the formulas used to derive the state-space matrices.

---

<sup>4</sup>Take into account, Quanser assumes  $\eta_g$  &  $\eta_m$  are equal to 1

The errors in Quanser's documentation are as follows:

- Incorrect inertia value reported for the rotary arm,  $J_r$ .
- Inconsistent assignment of the upper and lower link parameters in the MATLAB script.
- The efficiency parameters,  $\eta_g$  and  $\eta_m$ , are not accounted for in the provided model.
  - In this work, both efficiency parameters were set to unity. Future work could evaluate the effect of using the actual efficiency values on the model accuracy and controller performance.
- The motor resistance,  $R_m$ , appears to be counted twice in the back-EMF contribution to the calculation of the  $A$  matrix in the provided MATLAB script.
- The Maple derivation neglects damping friction, whereas the provided MATLAB implementation includes damping terms.

The updated table of values is as follows:

**Table 1:** DBIP Updated Parameter Values

| Parameter | Experimental (Correct) | Quanser's |
|-----------|------------------------|-----------|
| $L_r$     | 0.2159                 | 0.2159    |
| $L_1$     | 0.2000                 | 0.2000    |
| $l_1$     | 0.1635                 | 0.1635    |
| $L_2$     | 0.3365                 | 0.3365    |
| $l_2$     | 0.1778                 | 0.1778    |
| $m_1$     | 0.097                  | 0.097     |
| $m_h$     | 0.141                  | 0.1410    |
| $m_2$     | 0.127                  | 0.1270    |
| $J_r$     | 9.9829E-04             | 0.0041    |
| $\eta_g$  | 1                      | 0.90      |
| $\eta_m$  | 1                      | 0.69      |
| $K_g$     | 70                     | 70        |
| $k_t$     | 7.68E-3                | 7.68E-3   |
| $k_m$     | 7.68E-3                | 7.68E-3   |
| $R_m$     | 2.6                    | 2.6       |

Computing the AB matrices using the jacobian and evaluating at the point  $(x^*, u^*) = (x_{eq}, v_{eq})$  with:

$$A = \frac{\partial f}{\partial x} \Big|_{(x^*, u^*)} \quad B = \frac{\partial f}{\partial u} \Big|_{(x^*, u^*)} \quad (41)$$

results in matrices as follows:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

## 7 Controllability and Pole Placement

This section dealt with the stabilization of the DBIP about the upright linearization point. A system is controllable if the control input has sufficient authority to steer the system from any initial state to any desired state in finite time. The controllability of the linearized system was therefore verified by evaluating the rank of the following matrix:

$$\text{rank} \begin{bmatrix} B & AB & A^2B \cdots A^{n-1}B \end{bmatrix} \quad (42)$$

An example of a working set of desired eigenvalues is:

$$\sigma(A) = \{-109, -9 + 1.62j, -9 - 1.62j, -5, -4, -0.2\}$$

Using the place() function on MATLAB, the gain vector is obtained:

$$F = [-0.08989 \quad 11.082 \quad 40.481 \quad 0.037527 \quad 4.8218 \quad 5.1123]$$

## 8 Observability

This section addresses **state estimation**, since the physical system only provides direct measurements of the joint positions through the encoders. The corresponding joint velocities are obtained by differentiating the position measurements and filtering the resulting signals with a second-order-low-pass filter. This motivated the use of observers and the application of the procedure presented in [1, p. 125].

## 8.1 Full Order Observer Design - DBIP System

The observability of the pair  $(\mathbf{C}, \mathbf{A})$  was first verified by evaluating the rank of the observability matrix. Since the system was found to be observable, a full-order observer of the following form was implemented:

$$\begin{aligned}\dot{\hat{z}}(t) &= (\mathbf{A} + \mathbf{LC})\hat{z}(t) + (\mathbf{B} + \mathbf{LD})u(t) - \mathbf{L}y(t) \\ \hat{x}(t) &= \hat{z}(t)\end{aligned}\quad (43)$$

Where  $\mathbf{L}$  was picked such that  $\mathbf{A} + \mathbf{LC}$  was stable. The observer poles were selected as  $\sigma = \{-121, -122, -123, -124, -125, -126\}$  The inputs to the dynamic observer were:

$$\begin{bmatrix} u(t) \\ y(t) \end{bmatrix}$$

and will have the state space form

$$\left[ \begin{array}{c|cc} \mathbf{A} + \mathbf{LC} & \mathbf{B} + \mathbf{LD} & -\mathbf{L} \\ \hline \mathbf{I} & 0 & 0 \end{array} \right] \quad (44)$$

With the established poles, the full order observer was partitioned as:

$$\begin{aligned}A_o &= \begin{bmatrix} -172.32 & 1.5809 & -1.7248 & 1 & 0 & 0 \\ 87.911 & -243.25 & -3.973 & 0 & 1 & 0 \\ -83.815 & -3.7798 & -245.08 & 0 & 0 & 1 \\ -2677.6 & 280.35 & -294.98 & -74.816 & -1.7518 & 1.7906 \\ 14507 & -14666 & -636.14 & -82.886 & -2.1875 & 2.4444 \\ -13990 & -612.45 & -14763 & 84.725 & 2.4444 & -3.3478 \end{bmatrix} \\ B_o &= \begin{bmatrix} 0 & 172.32 & -1.5809 & 1.7248 \\ 0 & -87.911 & 243.25 & 3.973 \\ 0 & 83.815 & 3.7798 & 245.08 \\ 136.23 & 2677.6 & 213.41 & 291.39 \\ 150.92 & -14507 & 15263 & 612.43 \\ -154.27 & 13990 & 1.8385 & 14847 \end{bmatrix} \\ C_o &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ D_o &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\end{aligned}$$

## 8.2 Minimum Order Observer Design - DBIP System

The observability of the system is checked through the rank of the matrix:<sup>5</sup>:

$$\mathcal{O}(\text{observable}) \Leftrightarrow \text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^5 \end{bmatrix} = 6.$$

The pair (C,A) was observable, and the minimum-order observer was constructed as follows:

**Step 1.** Finding  $T^{-1}$

$$\text{Ker } \mathbf{C} = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

which represented the last  $n - p$  columns. The remaining basis vectors were then selected to complete a basis spanning  $\mathcal{X}$ , yielding

$$T^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

**Step 2.** Computed the following matrices and partitioned them:

$$TAT^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 \end{bmatrix}$$

$$TB = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \end{bmatrix} \quad CT^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

---

<sup>5</sup>This can be verified on MATLAB

**Step 3.** Picked  $L_2$  such that  $A_{22} + L_2A_{12}$  was stable. Using the `place()` function on MATLAB, the following Observer gains was obtained with the following desired poles

$$\sigma_{observer} = \{-73, -74, -75\}$$

$$L_2 = \begin{bmatrix} 1.8161 & 1.7518 & -1.7906 \\ 82.886 & -71.812 & -2.4444 \\ -84.725 & -2.4444 & -71.652 \end{bmatrix}$$

**Step 4.** The minimal order observer was

$$\dot{z}(t) = Mz(t) + Nu(t) + Py(t)$$

$$\hat{x}(t) = Qz(t) + Ru(t) + Sy(t)$$

where

$$M := A_{22} + L_2A_{12}, \quad N := B_2 + L_2B_1, \quad P := (A_{21} + L_2A_{11} - A_{22}L_2 - L_2A_{12}L_2)C_1^{-1},$$

$$Q := T^{-1} \begin{bmatrix} 0 \\ I_{n-p} \end{bmatrix}, \quad R := 0, \quad S := T^{-1} \begin{bmatrix} C_1^{-1} \\ -L_2C_1^{-1} \end{bmatrix}.$$

giving

$$M = \begin{bmatrix} -73 & 0 & 0 \\ 0 & -74 & 0 \\ 0 & 0 & -75 \end{bmatrix} \quad N = \begin{bmatrix} 136.23 \\ 150.92 \\ -154.27 \end{bmatrix} \quad P = \begin{bmatrix} 132.57 & 621.64 & -134.3 \\ 6133.6 & -4716.8 & -204.59 \\ -6354.4 & -793.94 & -5290.5 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad S = \begin{bmatrix} 1.0000 & 0 & 0 \\ 0 & 1.0000 & 0 \\ 0 & 0 & 1.0000 \\ -54.8161 & -1.7518 & 1.7906 \\ -82.8864 & 22.8125 & 2.4444 \\ 84.7249 & 2.4444 & 26.6522 \end{bmatrix}$$



To improve the convergence of the state estimates, the initial condition  $z(0)$  was selected to correspond to zero initial velocity at the three joints near the upright position. In the Quanser implementation, the controller was engaged once both  $\alpha$  and  $\phi$  were within  $1.5^\circ$  of the upright equilibrium. From [1, p. 124], it's given that:

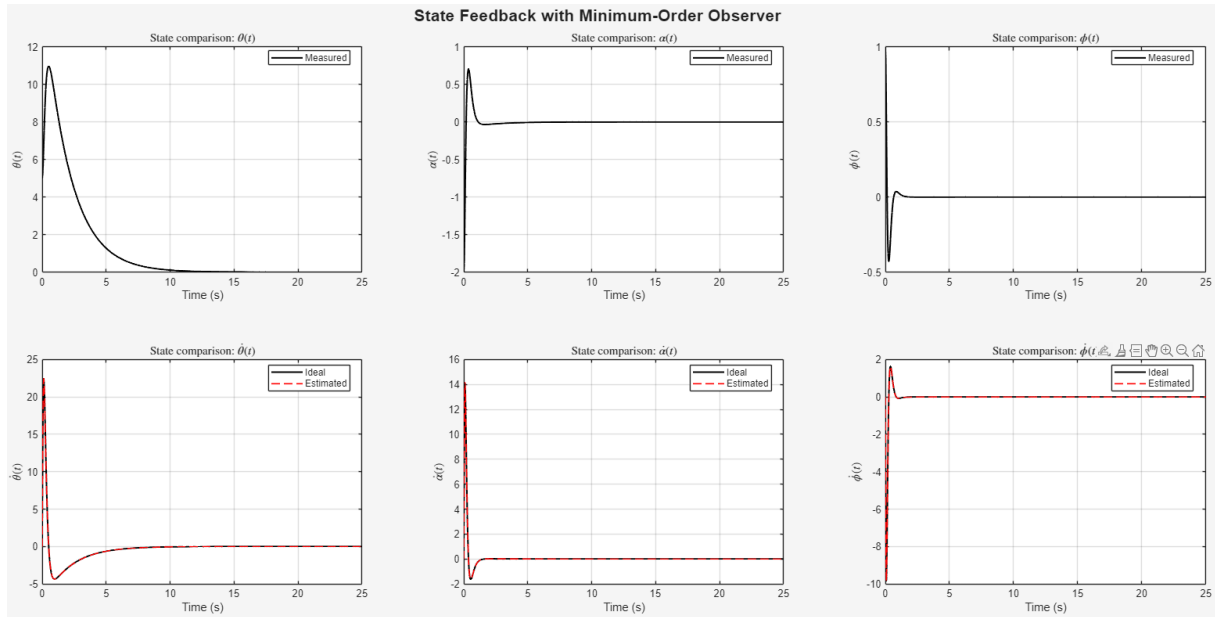
$$z(t) := \hat{x}_2(t) + L_2 C_1^{-1} y(t) \quad (45)$$

Since  $\hat{x}_2(t)$  is 0 at the near upright position, once the control engages, the initial condition was reset to

$$z(0) = L_2 C_1^{-1} y(0) \quad (46)$$

Where  $y(0)$  is the position measurement at the time the control is engaged. Using the state-feedback gains from 8, the following simulated response for an arbitrary initial condition was obtained:

$$x_o = [5 \quad -2 \quad 1 \quad 0 \quad 0 \quad 0]^T$$



**Figure 2:** Minimum Order Observer with State Feedback Controller

### 8.3 Hardware Implementation

An unit delay block was introduced to the input of the observer because the build of the file created an algebraic loop.

## 9 Regulation and Tracking

This section deals with the tracking of a step reference signal and the regulation of the plant. The reference signals are assumed to satisfy known differential equations.

### 9.1 The Case of Full Information

Assuming full state feedback, the case of full information was implemented<sup>6</sup> for tracking. The system consisted of a plant with state  $x_1$  and an exosystem with state  $x_2$  as follows:

$$\begin{aligned} \dot{x}_1 &= A_1 x_1 + A_3 x_2 + B_1 u \\ \dot{x}_2 &= A_2 x_2 \\ e &= D_1 x_1 + D_2 x_2 \\ y &= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned} \quad (47)$$

Combining the two states

$$\dot{x} = Ax + Bu, \quad e = Dx, \quad y = x, \quad (48)$$

$$A = \begin{bmatrix} A_1 & A_3 \\ 0 & A_2 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, \quad D = \begin{bmatrix} D_1 & D_2 \end{bmatrix}.$$

and with the controlled system ( $u = Fx$ )

$$\dot{x} = (A + BF)x, \quad e = Dx, \quad (49)$$

$$A + BF = \begin{bmatrix} A_1 & A_3 \\ 0 & A_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} \begin{bmatrix} F_1 & F_2 \end{bmatrix} = \begin{bmatrix} A_1 + B_1 F_1 & A_3 + B_1 F_2 \\ 0 & A_2 \end{bmatrix}.$$

The regulation problem is one that solves the condition that  $e(t)$  goes to zero for every  $x(0)$ . Thus, the **regulator equations** were solved to determine whether the regulation problem was solvable using a control law of the form  $u = Fx$ .

Here the state variables of the plant and the exomodel are:

$$x_1 := \begin{bmatrix} \theta \\ \alpha \\ \phi \\ \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \end{bmatrix} \quad x_2 := \begin{bmatrix} r \end{bmatrix} \quad (50)$$

---

<sup>6</sup>This method doesn't work very well experimentally

The regulated variable was defined as the tracking error

$$e := r - \theta \quad (51)$$

The controller had access to the full state of the pendulum and the reference signal

$$y := \begin{bmatrix} \theta \\ \alpha \\ \phi \\ \dot{\theta} \\ \dot{\alpha} \\ \dot{\phi} \\ r \end{bmatrix} \quad (52)$$

The resulting system was expressed in the following form:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 & 0 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 & 0 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (53)$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \\ 0 \end{bmatrix} \quad (54)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (55)$$

$$D = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (56)$$

Next,  $F_1$  was selected such that  $A_1 + B_1 F_1$  is stable as discussed in section 8

$$F_1 = \begin{bmatrix} -0.4870 & 24.6576 & 60.4800 & -0.6999 & 8.1836 & 7.7908 \end{bmatrix}$$

Next, the following equations:

$$A_1X - XA_2 + A_3 + B_1U = 0, \quad D_1X + D_2 = 0 \quad (57)$$

for  $X$  and  $U$ . Looking at the augmented matrices, it's observed that

$$A_1 \in \mathbb{R}^{6 \times 6}, \quad A_2 \in \mathbb{R}^1, \quad A_3 \in \mathbb{R}^{6 \times 1} \quad (58)$$

and concluded that

$$X = \begin{bmatrix} x_{11} \\ x_{21} \\ x_{31} \\ x_{41} \\ x_{51} \\ x_{61} \end{bmatrix}, \quad U = [u_{11}] \quad (59)$$

where

$$X = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad U = [0] \quad (60)$$

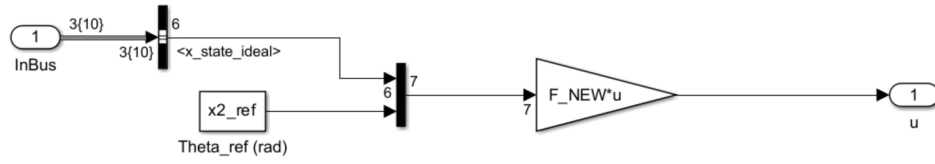
resulting in the state feedback controller:

$$\begin{aligned} F &= [F_1 \quad F_2] = [F_1 \quad U - F_1X] \\ &= [-0.48705 \quad 24.658 \quad 60.48 \quad -0.69993 \quad 8.1836 \quad 7.7908 \quad 0.48705] \end{aligned} \quad (61)$$

The first 6 gains are for the plant state variables and the last entry  $F_2$  is for the step reference tracking. The exosystem is represented by

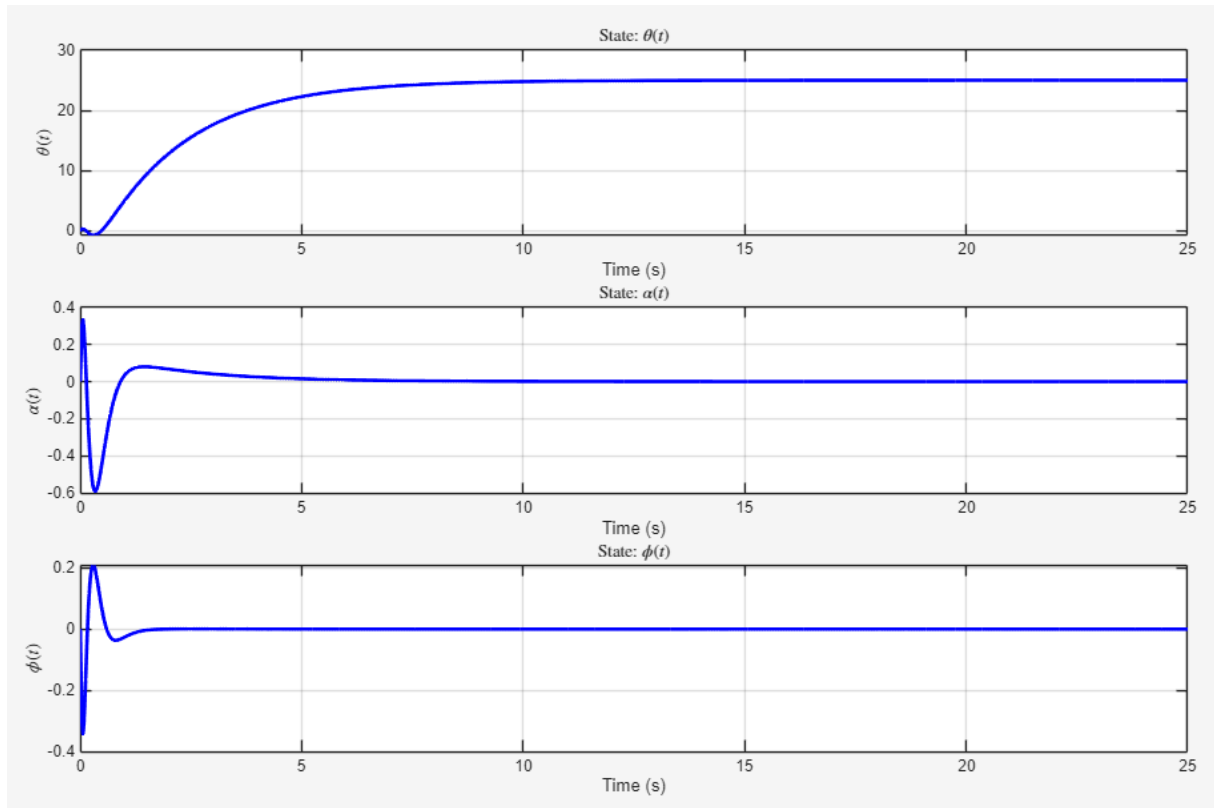
$$\dot{x}_2 = A_2x_2 \quad (62)$$

Where  $A_2$  is zero,  $B = 0$ ,  $C = 1$ , and  $D = 0$ , giving the constant output  $x_2$ . The latter can be represented using a constant block and ensuring the value is converted into radians before being combined with  $x_1$ .



**Figure 3:** Reference Tracking Controller - Case of Full Information

With the latter controller, a demonstration of reference tracking is shown in Figure 4



**Figure 4:** Reference Tracking Plot - Desired  $\theta = 25$

## 9.2 The Case of Measurement Feedback

This section<sup>7</sup> considers the case where the controller does not have sensors for the full state  $x_1$  and  $x_2$  of the exosystem. The augmented plant is similarly of the form.

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 & 0 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 & 0 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (63)$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \\ \hline 0 \end{bmatrix} \quad (64)$$

with

$$y = C_1 x_1 + C_2 x_2 \quad (65)$$

$$C_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad C_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

and

$$e = D_1 x_1 + D_2 x_2 \quad (66)$$

$$D_1 = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad D_2 = \begin{bmatrix} 1 \end{bmatrix}$$

$F$  was chosen to be the state feedback using the same methodology as in the previous subsection, resulting in

$$F = \begin{bmatrix} -0.33398 & 18.351 & 51.673 & -0.332 & 6.647 & 6.5995 & 0.33398 \end{bmatrix} \quad (67)$$

An  $L$  was chosen such that  $A + LC$  remained stable, and constructed the observer-based controller with  $\sigma(A + LC) = \{-161, -162, -163, -164, -165, -166, -167\}$

$$\dot{x}_c(t) = (A + BF + LC)x_c(t) - Ly(t) \quad (68)$$

$$u(t) = Fx_c(t).$$

---

<sup>7</sup>This method doesn't work very well experimentally

with

$$L = \begin{bmatrix} -253.78 & 1.62 & -1.1372 & 0 \\ 82.735 & -327.74 & -4.3533 & 7.1054 \times 10^{-15} \\ -83.64 & -3.7341 & -325.13 & -7.1054 \times 10^{-15} \\ -8300 & -69.142 & -378.61 & -4.5475 \times 10^{-13} \\ 20625 & -27235 & -981.73 & 0 \\ -20842 & -253.28 & -26075 & -1.819 \times 10^{-12} \\ -161 & 0 & 0 & -161 \end{bmatrix} \quad (69)$$

### 9.3 Structurally Stable Synthesis

This section deals with constructing a structurally stable synthesis. The controller has a state vector with two components  $x_c = (x_{c1}, x_{c2})$  and the controller has the form

$$\begin{bmatrix} \dot{x}_{c1} \\ \dot{x}_{c2} \end{bmatrix} = \begin{bmatrix} A_{c1} & 0 \\ A_{c3} & A_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + \begin{bmatrix} B_{c1} \\ B_{c2} \end{bmatrix} y \quad (70)$$

$$u = \begin{bmatrix} C_{c1} & C_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + D_c y$$

with  $y = (y_o, e)$ . Next:

$$C_1 = \begin{bmatrix} C_{01} \\ D_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad C_2 = \begin{bmatrix} C_{02} \\ D_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (71)$$

giving

$$y = \begin{bmatrix} y_o \\ e \end{bmatrix} = \begin{bmatrix} \theta \\ \alpha \\ \phi \\ r - \theta \end{bmatrix} \quad (72)$$

The procedure for the construction of the controller is as follows:

**Step 1:** Denote the characteristic polynomial of  $A_2$  (exomodel) by

$$\pi_2(s) = \det(sI - A_2) = \det(s - 0) = s \quad (73)$$

where  $n_2 = 1$ ,  $\alpha_0 = 0$  and  $q = 1$  (number of regulated variables. Set

$$A_{c1} = [0] \quad (74)$$

$$B_{ce1} = [1]$$

and set

$$B_{c1} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \quad (75)$$

**Step 2:** Form the **augmented system** consisting of the plant and the portion of the controller from Step 1

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_{c1} \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ B_{c1}C_1 & A_{c1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u$$

$$y_a = \begin{bmatrix} C_1 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix}$$

therefore

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_{c1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 493.7629 & -3.5865 & -74.8161 & -1.7518 & 1.7906 & 0 \\ 0 & 597.3631 & -23.7073 & -82.8864 & -2.1875 & 2.4444 & 0 \\ 0 & -610.6139 & 83.3820 & 84.7249 & 2.4444 & -3.3478 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 136.2256 \\ 150.9200 \\ -154.2677 \\ 0 \end{bmatrix} u$$

$$y_a = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_{c1} \end{bmatrix}$$

To design a stabilizer for the augmented system

$$F_a = \begin{bmatrix} F_{a1} & F_{a2} \end{bmatrix}$$

was chosen so that  $A_a + B_a F_a$  is stable with the poles  $\sigma = \{-110, -9.85 + 1.62i, -9.85 - 1.62i, -6, -4.2, -0.5 + 0.27i, -0.5 - 0.27i\}$ . An  $L_1$  was chosen making  $A_1 + L_1 C_1$  stable with poles placed at  $\sigma = \{-147, -149, -151, -153, -155, -157\}$  resulting in

$$L_1 = \begin{bmatrix} -114.57 & 1.55 & -1.61 & 114.57 \\ 44.67 & -298.51 & -4.64 & -44.67 \\ -40.78 & -4.11 & -304.01 & 40.78 \\ -3121.66 & -122.95 & -385.86 & 3121.66 \\ 9790.12 & -22680.28 & -908.40 & -9790.12 \\ -9217.63 & -224.96 & -22823.40 & 9217.63 \end{bmatrix} \quad (76)$$

Thus the controller has the form

$$\begin{bmatrix} \dot{x}_{c1} \\ \dot{x}_{c2} \end{bmatrix} = \begin{bmatrix} A_{c1} & 0 \\ A_{c3} & A_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + \begin{bmatrix} B_{c1} \\ B_{c2} \end{bmatrix} y \quad (77)$$

$$u = \begin{bmatrix} C_{c1} & C_{c2} \end{bmatrix} \begin{bmatrix} x_{c1} \\ x_{c2} \end{bmatrix} + D_c y$$



And replacing it with the numeric values previously obtained

$$\begin{bmatrix} A_{c1} & 0 \\ A_{c3} & A_{c2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -229.13 & 1.55 & -1.61 & 1.00 & 0 & 0 \\ 0 & 89.34 & -298.51 & -4.64 & 0.00 & 1.00 & 0 \\ 0 & -81.57 & -4.11 & -304.01 & 0 & 0 & 1.00 \\ 29.96 & -6354.94 & 3344.70 & 7158.40 & -156.87 & 1009.97 & 972.57 \\ 33.19 & 19456.56 & -18788.24 & 7429.91 & -173.79 & 1118.67 & 1077.94 \\ -33.92 & -18308.84 & -4203.32 & -31287.53 & 177.64 & -1143.27 & -1102.70 \end{bmatrix}$$

$$\begin{bmatrix} B_{c1} \\ B_{c2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1.00 \\ 114.57 & -1.55 & 1.61 & -114.57 \\ -44.67 & 298.51 & 4.64 & 44.67 \\ 40.78 & 4.11 & 304.01 & -40.78 \\ 3121.66 & 122.95 & 385.86 & -3121.66 \\ -9790.12 & 22680.28 & 908.40 & 9790.12 \\ 9217.63 & 224.96 & 22823.40 & -9217.63 \end{bmatrix}$$

$$\begin{bmatrix} C_{c1} & C_{c2} \end{bmatrix} = \begin{bmatrix} 0.2199 & -0.8194 & 21.8306 & 55.4070 & -0.6023 & 7.4268 & 7.1263 \end{bmatrix}$$

$$D_c = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

## 10 Next steps

The work presented in this report encompassed a substantial portion of the multivariable control systems material taught at the University of Waterloo. Possible directions for future work include:

- Investigation of nonlinear control methods.
- Verification of the motor and gearbox efficiency parameters used in the commanded voltage model, and evaluation of their effect on plant performance.
- Development of a swing-up controller.
- Comparison of additional control strategies, such as robust control or model predictive control.

## References

- [1] Dr. Christopher Nielsen. Multivariable control systems. *Waterloo University*, 2026. pages 13, 17
- [2] {Mark W.} Spong, Seth Hutchinson, and M. Vidyasagar. Robot modeling and control. *IEEE Control Systems*, 26(6):113–115, December 2006. pages 8